Genuine Multipartite Entanglement Analysis

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Model                              Ground-state properties           GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E0                : -1.27491722  Extracted n-spin RDM : n = 3
                                   E1                : -1.27491722
System parameters                                                   Purity:
-----------------                  <S2>              : 0.750000       gamma = Tr(rho^2)
Sites (L)          : 3             S                 : 0.5000
Electrons (Ne)     : 3             <Sz>              : 0.250000     Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1 : 2.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state        : Doublet (S=1/2)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 3             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total         : 0.230835     Convex roof (mixed):
                                   <D> / site        : 0.076945       C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ))

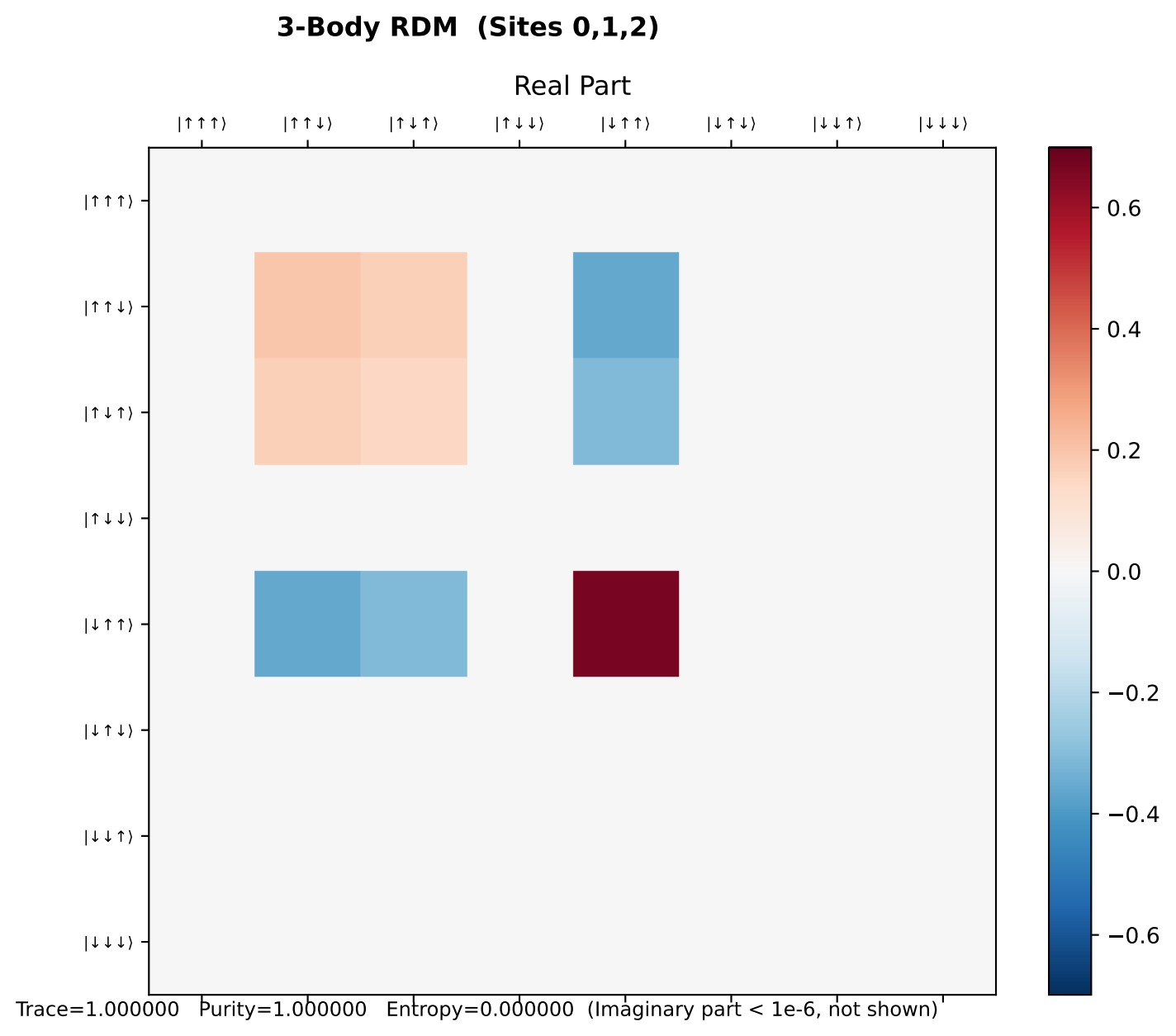

                                                                    Date: 2026-04-10  21:07
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Summary: All 3-Body RDM Clusters

System: L=3, Ne=3, U=4.0, t=1.0 | E_0=-1.274917

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_UB	Status
(0, 1, 2)	1.000000	0.700793	0.000000	0.700793	0.700793	0.700793	GME
(1, 2, 0)	1.000000	0.700793	0.000000	0.700793	0.700793	0.700793	GME
(2, 0, 1)	1.000000	0.700793	0.000000	0.700793	0.700793	0.700793	GME

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.673436e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (0,)         | (1, 2)       | 0.55479385
  (0, 1)       | (2,)         | 0.69076186
  (0, 2)       | (1,)         | 0.75444429

STATUS: GME EXISTS
------------------------------------

C_GME = 0.70079342
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.70079342
  C_LB    = 0.00000000
  Gap     = 0.70079342
```

```
Convex Roof Analysis
====================================

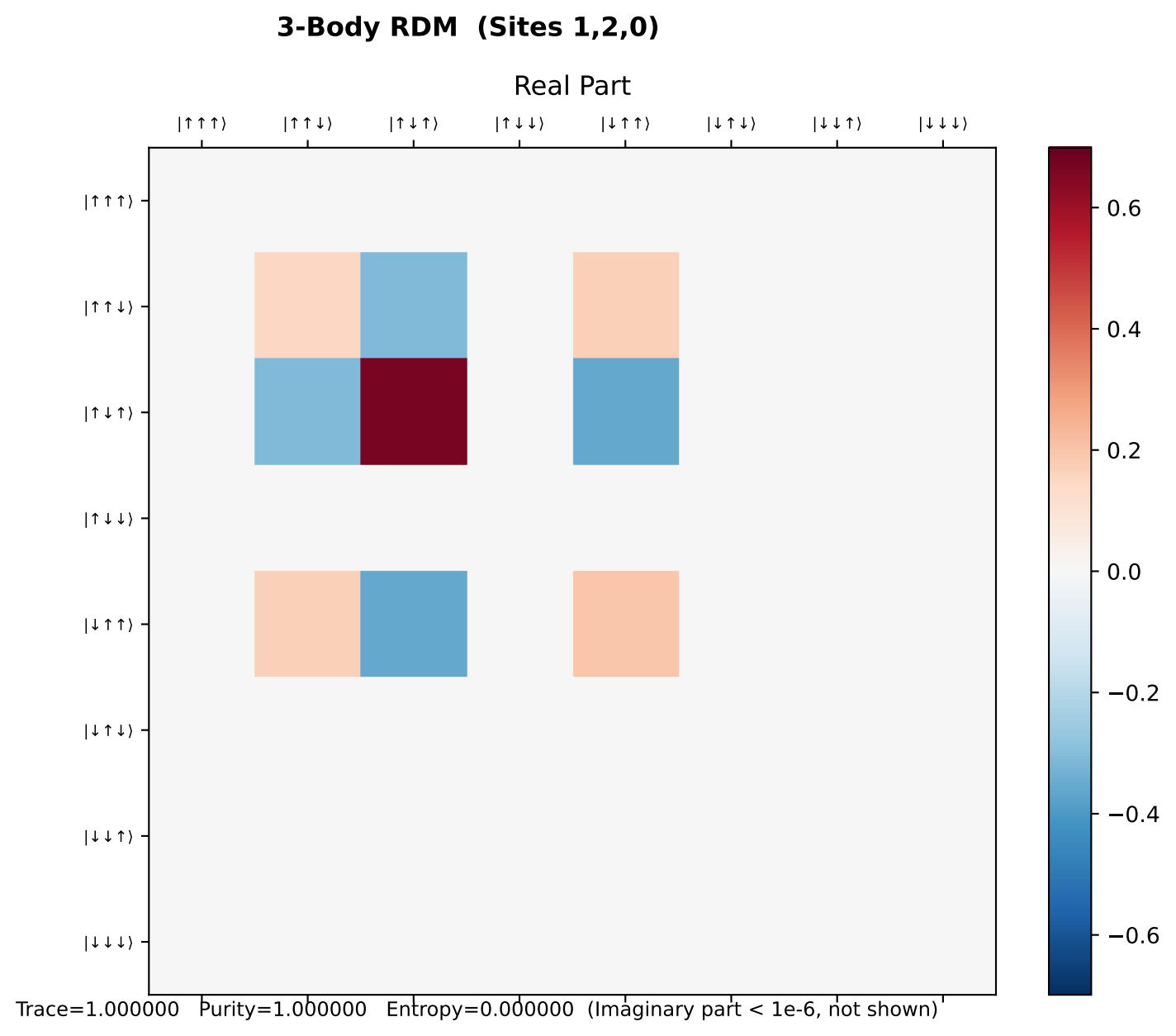
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.70079342   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.70079342   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.70079342

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


3-Body RDM (Sites 1,2,0)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -3.112091e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (1,)         | (2, 0)       | 0.69076186
  (1, 2)       | (0,)         | 0.75444429
  (1, 0)       | (2,)         | 0.55479385

STATUS: GME EXISTS
------------------------------------

C_GME = 0.70079342
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.70079342
  C_LB    = 0.00000000
  Gap     = 0.70079342
```

```
Convex Roof Analysis
====================================

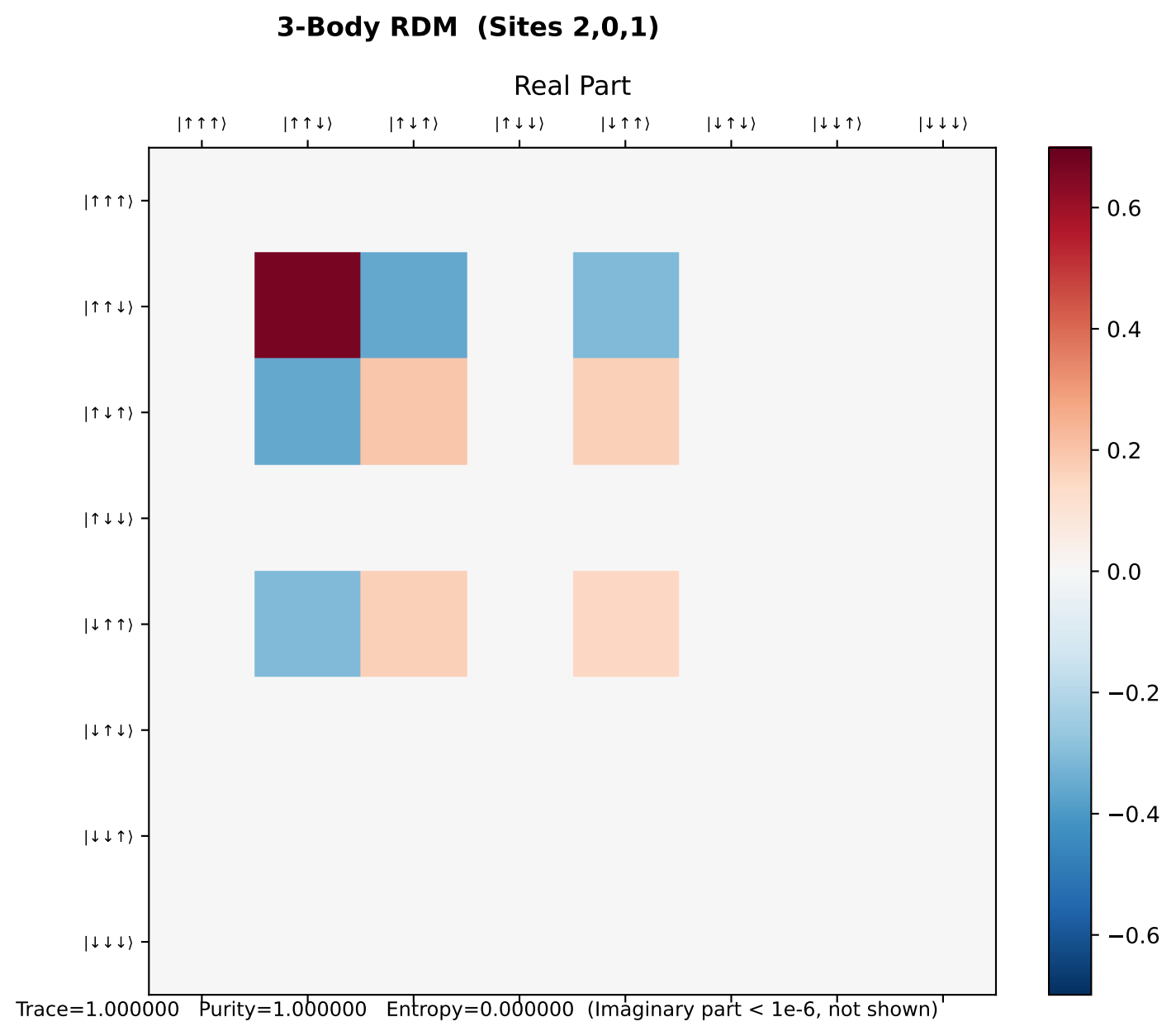
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.70079342   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.70079342   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.70079342

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


3-Body RDM (Sites 2,0,1)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,0,1)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.154627e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (2,)         | (0, 1)       | 0.75444429
  (2, 0)       | (1,)         | 0.55479385
  (2, 1)       | (0,)         | 0.69076186

STATUS: GME EXISTS
------------------------------------

C_GME = 0.70079342
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.70079342
  C_LB    = 0.00000000
  Gap     = 0.70079342
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.70079342   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.70079342   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.70079342

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```